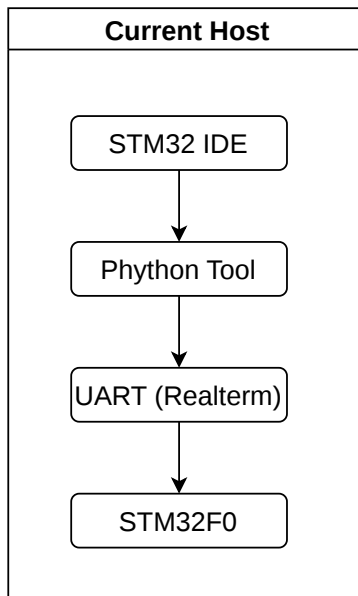


Design Requirements

Purpose
• Bare-metal • Dual-slot • STM32F0 • Foundation for H7

Functional
• UART Update • CRC16 • Metadata • Slot Manager • Recovery

Boot Policy
• Wait 1 sec • Boot if valid • Retry ×5 • Wait forever if invalid



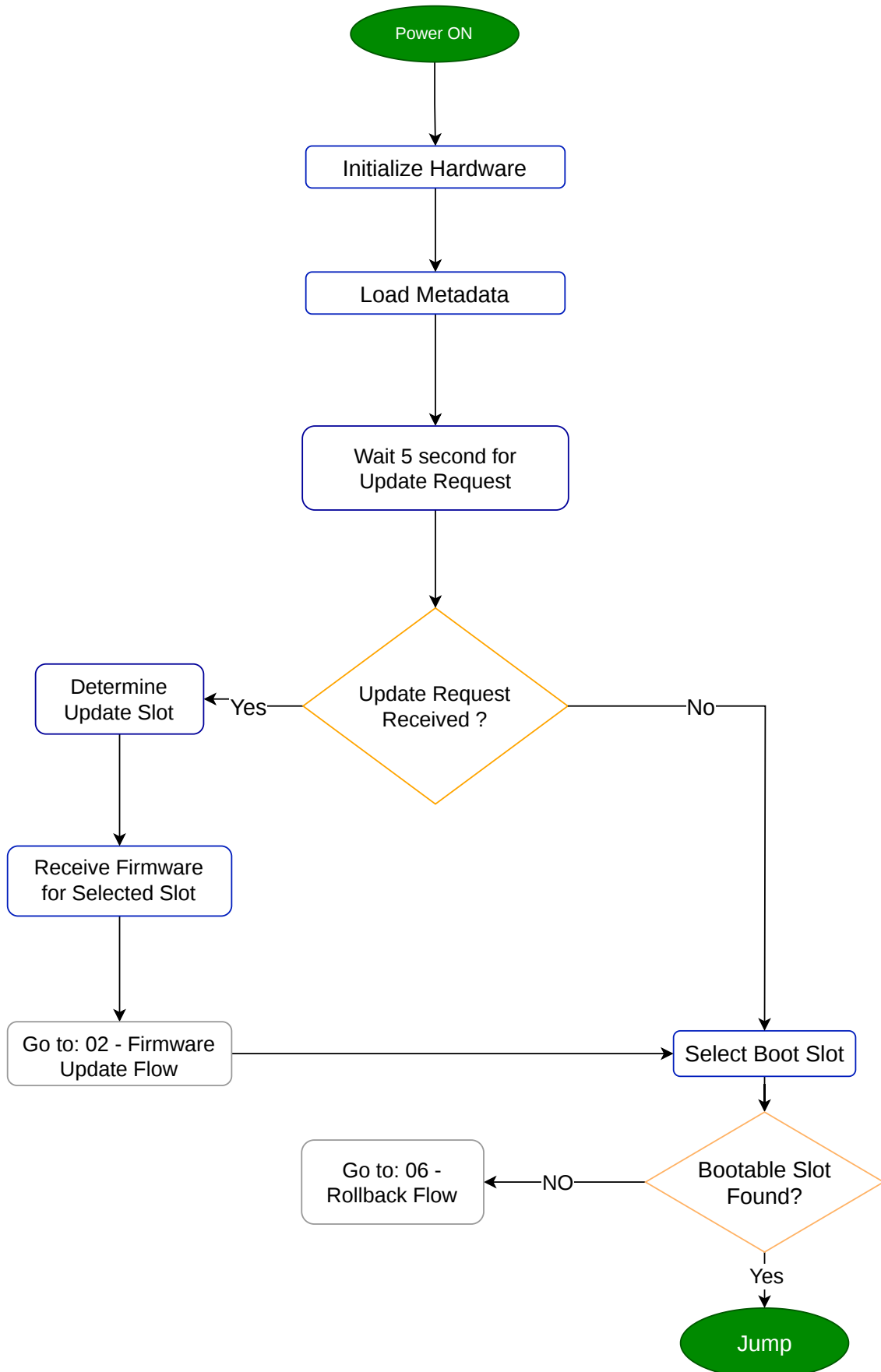
Future
• STM32H7 • ESP32 • BLE • Wi-Fi • SHA-256

Status

- Architecture : Frozen
- Version : v1.0
- Last Updated : 12 - 07 - 2026

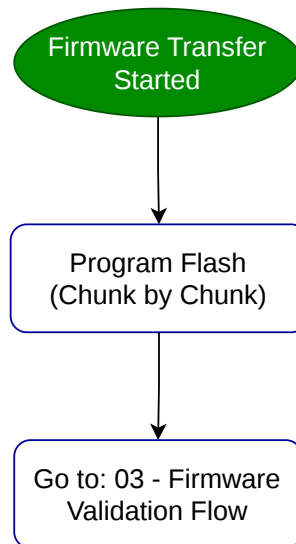
Power-on Boot Flow

Question - What happens after reset?



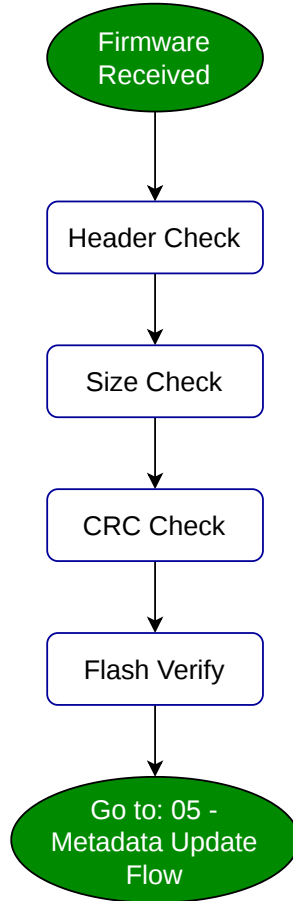
Firmware Update Flow

How does the new firmware get installed



Firmware Validation Flow

How is the new firmware validated?

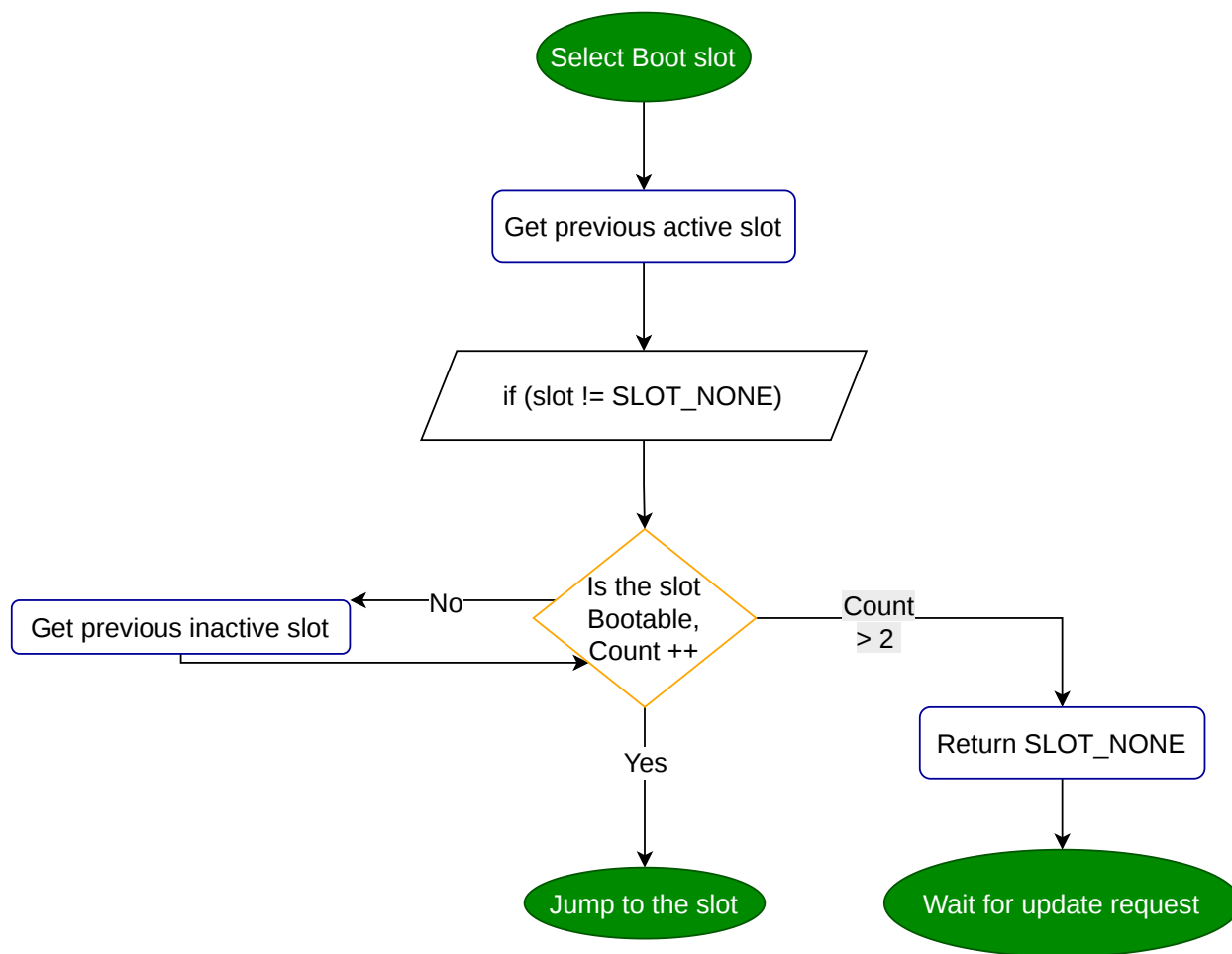


Note :

If any validation step fails, the firmware update is rejected and control returns to the Firmware update flow.

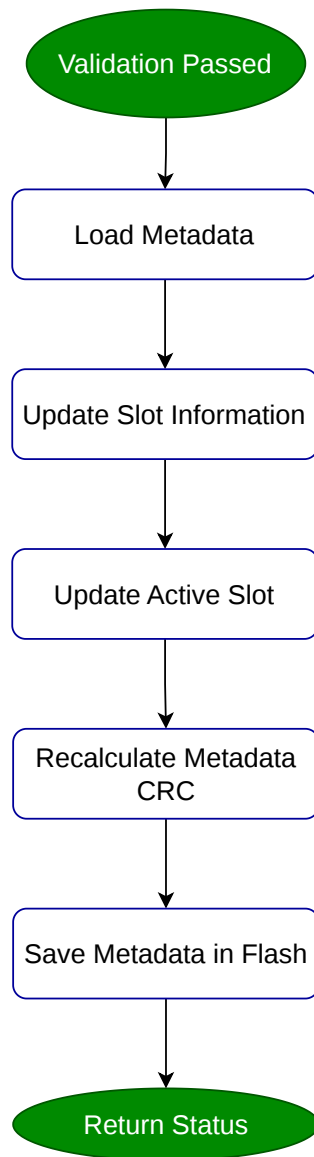
Slot Selection Flow

Which Firmware should execute



Metadata Update Flow

when and how is metadata modified

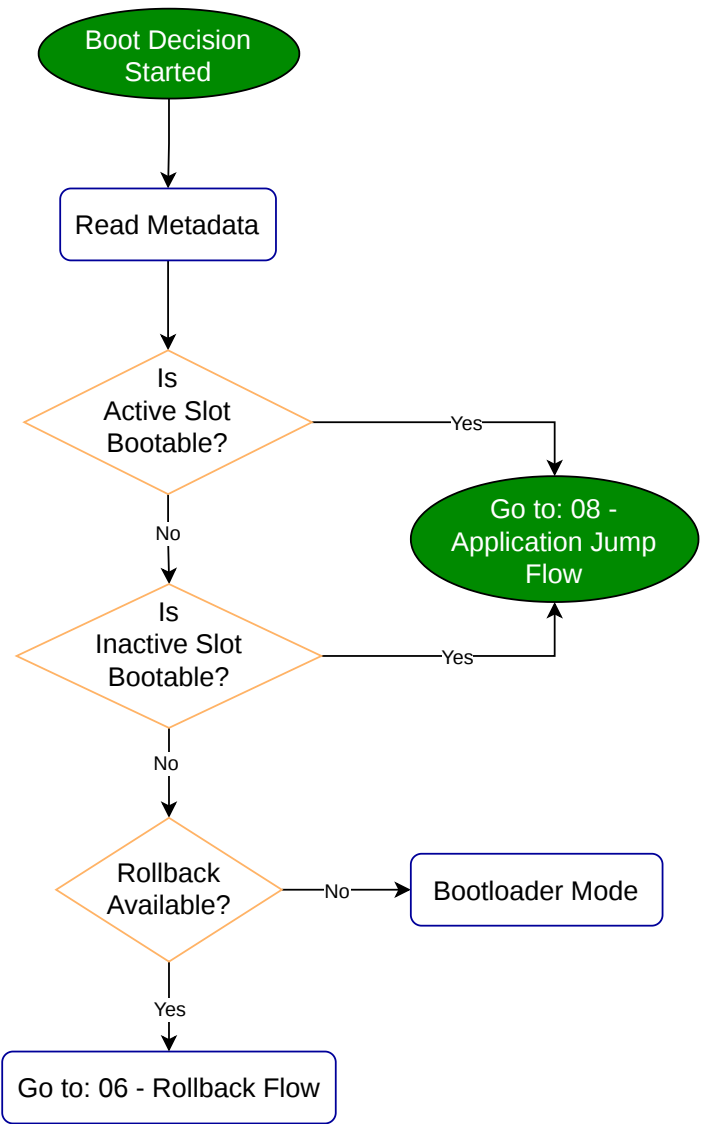


Rollback Flow

How does the system recover from a failed updates

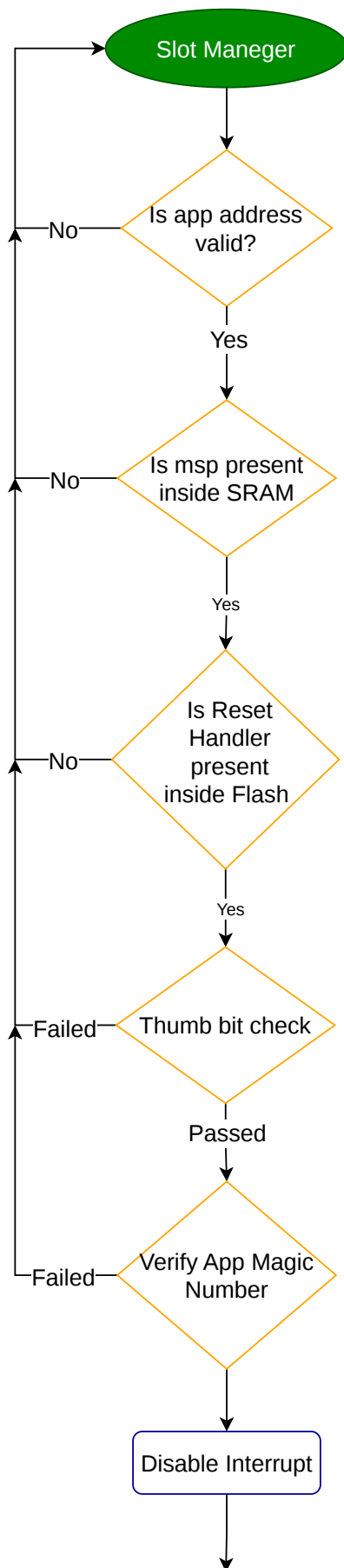
Boot Slot Selection Flow

which slot to jump?



Application Jump Flow

How is control transferred safely to the application



Relocate Vector Table
from app address to
SRAM



Set MSP to app
address



Jump to Application
Address